Competition 1: The LLM Safety Mandate

Course: IFT 3395 / IFT 6390A — Fundamentals of Machine Learning
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1. Exploration

The exploration begins with Data (1,800 train and 1,198 test rows), five text features(modal_conditionnel, mise_a_distance, imperatif_debut, n_obligations, part_majuscules) and four unknown variables v1 to v4. The target refus is 1 for 45% of rows, so always predicting 0 scores for 55% of rest. There are no missing values or duplicates, as well as train and test match closely, so we did not expect a distribution shift from the data. The five named features are weakly related to the target.

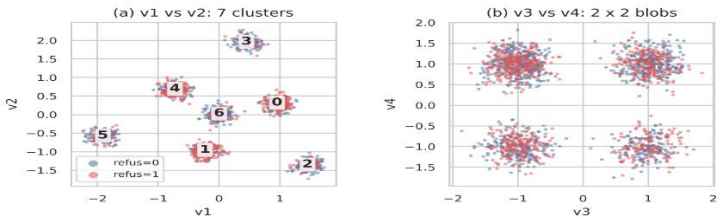


Figure 1. Exploration of the anonymized variables, colored by class (a, b)

This pair plot was the most informative view shown in figure 1. The (v1, v2) form 7 tight clusters and v3, v4 each split into two blobs at about -1 and +1.

What this told me about which methods to try?

- Linear models on the raw features should be weak because the correlations and the structure(a logistic regression on the raw features confirmed this, at 62.5% accuracy).
- Treat the clusters as categories and look for interactions with v3 and v4, as classes overlap inside each cluster.
- Compare interaction-capable models (forests, boosting) with kNN, SVM and Neural networks.
- Keep models simple and validation repeated because n is small and the classes overlap.

2. The anonymized features

I used all four, plus the five named features. In Figure 2, v3 and v4 alone score 53.4% and the flag "both positive" alone 54.9%, while v1 and v2 alone reach 65.0%. Together they follow an XOR-style rule, refus is about [cluster in {0, 1, 4}] XOR [v3 > 0 and v4 > 0], which gives 89.1%, against 82.8% for boosted trees on the raw v1 to v4. Keeping named features alone reach 70.7% and add AUC (0.896 to 0.906) but no change in accuracy.

What could they represent?

The (v1, v2) clusters are 0.12 SD wide and at least 0.99 apart, so v1 and v2 behave like a 7-level categorical variable. Three named features (imperatif_debut, part_majuscules, modal_conditionnel) differ between high- and low-refusal clusters (Mann-Whitney $p < 0.01$), so the clusters relate weakly.

The hypotheses: (v1, v2) as a category of request, (v1, v2) as an identity as refusal behaviour differs and v3, v4 as binary context flags.

Why would a company hide them?

Privacy (embeddings and metadata derived from user prompts can leak personal content), security (our own result shows that knowing them predicts refusals with 89% accuracy that avoid refusals), and trade secrets.

Is it acceptable to use features without knowing what they measure?

For this task, the goal is prediction and the validation shows the relationship is robust, so using the variables is justified. Before any real use, we would require a data dictionary at least for auditors, a check for proxy relationships with protected attributes, and monitoring of the clusters and flags distributions for drift. Otherwise, all interpretations will remain just as hypotheses.

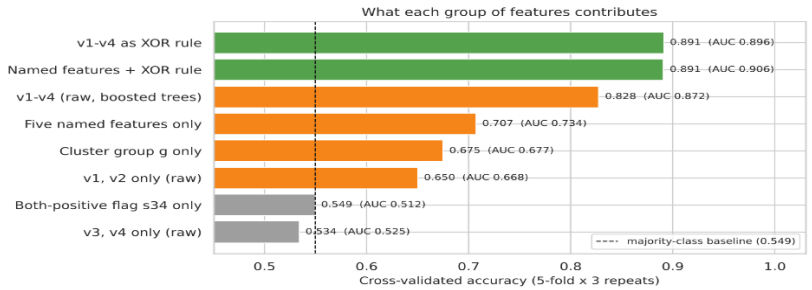


Figure 2. Cross-validated accuracy for different sets of features.

3. Methodology

Split: We used stratified 5-fold cross-validation repeated 3 times on the training file (15 fits, fixed seeds) and report mean and SD. Everything that uses labels (the cluster-to-group mapping behind the XOR feature) is relearned inside each training fold; KMeans (k = 7, best silhouette, 0.81, for k = 3 to 10) uses no labels and is fitted on the v1, v2 values of train and test.

Models: Six on the raw features (logistic regression, kNN, SVM-RBF, random forest, histogram boosting, MLP) and five on engineered features (the rule only, logistic regression, shallow boosting, random forest, logistic regression on smoothed segment rates), then a soft-voting ensemble of logistic regression, boosting and forest. All models are shown in Figure 3.

Metrics: accuracy (primary), AUC, log loss.

Hyperparameters: Set a priori to conservative values (strong regularization, depth-2 boosting, minimum leaf 15 to 40. A sensitivity checks afterwards over 20 settings gave 0.8906 to 0.8913, except a forest with leaf size 1 (0.862).

Threshold: The value of 0.5. A sweep from 0.20 to 0.80 on out-of-fold probabilities left accuracy (0.8911) and F1 (0.8864) flat from 0.25 to 0.60, so 0.5 sits mid-plateau (no test row lies within 0.185 of it).

4. Results and discussion

The best raw-feature model is the random forest (84.2%) from every engineered model and the final ensemble reach 89.1% (SD 1.9 points, AUC 0.906, F1 0.886).

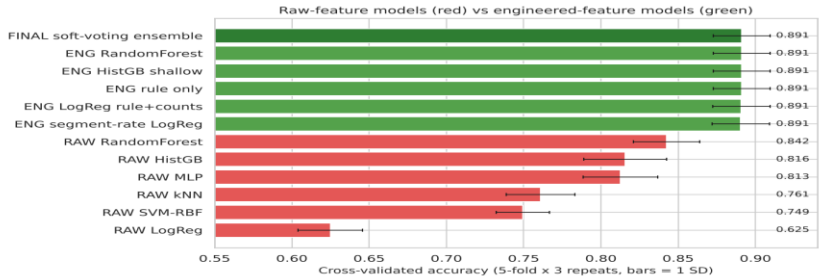


Figure 3. Cross-validated accuracy of all models

I think, what limits the score is label noise, not model. The rule misclassifies about 11% of training rows, and I found no way to predict which ones. It barely reaches AUC 0.655, and errors are flat across every feature and the id. A forest that tries to fit them overfits (0.862).

One limitation of the dataset: About 11% of the labels behave like irreducible noise which gives no idea of their origin, because the variables that might explain it are anonymized. This results accuracy near 89% for the methods which I tried.

5. AI disclosure

I used Claude and ChatGPT to understand the data more deeply and generate figures of the data as well as models. I used them to get the information regarding different models to try on this type of data and learn about engineered features technique and knowing the default threshold values used. I also used to clean the code and solve the errors. Additionally, I had done data visualization by myself and wright the code by myself at first instance along with the comments, I just used AI for my assistance and learn more things about models and data which I do not know.